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## PLANAR TRANSFORMER AND DUAL ACTIVE BRIDGE


### Abstract

A protective enclosure for electrical components includes potting material encasing the electrical components. A case covers the potting material. At least one clastic component extends over the case for applying a compressive load to the potting material.



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### Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of U.S. application Ser. No. 18/585,173, filed Feb. 23, 2024, the entirety of which is incorporated by reference herein.

## TECHNICAL FIELD

[0002] The present invention relates generally to transformers, and specifically to a planar, high power, medium voltage transformer and an associated dual-active bridge module.

## BACKGROUND

[0003] Medium voltage AC to DC converters can be used in electric vehicle (EV) charging stations. More specifically, medium-voltage AC to ˜1000 VDC conversion is accomplished using many identical, high-frequency transformer-based AC to DC conversion modules connected in series on the AC medium voltage side and connected in parallel on the DC side-a so-called series-in parallel-out architecture. By placing the primary windings in series and the secondary windings in parallel, a large voltage step down can be achieved with transformers that are 1:1 or nearly 1:1 turns ratio. The transformers can achieve a frequency of about 20 kHz-200 kHz.

[0004] Due to this construction, however, the eddy current and AC losses can be high. As a result, the conductors in the transformer tend to be made from LITZ wire, laminated tape or arranged in a planar transformer. LITZ wire has fine strands of conductors that are insulated from each other but can be expensive. Moreover, the insulation used with LITZ wire tends to occupy a large percentage of the winding space. Laminated tape is thin and provides wide layers of conductors separated by insulation, which is difficult to implement in standard cores because edge winding is problematic with laminated tape.

[0005] A second challenge is the high voltage isolation that is needed between the primary and secondary windings in the transformer. Because the primary windings are many different voltages and the secondary voltages are similar, significant isolation is required. This can require 30-50 kV isolation for 15 kV class systems. Further, these devices require a Basic Impulse Level (BIL) of over 100 kV. These isolation voltages require either very large creepage and clearance distances or continuous solid insulation between the layers. Additionally, there is a desire to have the conductive components for transformer cores and for cooling to be ground referenced and isolated from any of the winding voltages.

[0006] A third challenge is the minimization of parasitic resistances, capacitances, and inductances in locations that make it difficult to switch at high frequencies and that contribute to heat generation that needs to be removed and reduce efficiencies. While some inductance is necessary for Zero Voltage Switching (ZVS) operation, too much results in excess reactive current in the bus capacitors, as well as additional losses in snubbers. Capacitance within the winding makes it more challenging to achieve ZVS switching and capacitance across the ground wall increases losses in the switches, and contributes to EMI issues.

[0007] A fourth challenge is to keep the transformer cool while achieving the above three requirements. If liquid cooling or heat pipes are employed, placing metal in or near the transformer can have high eddy current losses depending on how it is arranged, how conductive the metal is, and how thick and what orientation the metal placed. Further, the transistors added on each side of the transformer have similar requirements of low parasitics, good cooling, and high isolation.

## SUMMARY

[0008] A high-frequency planar transformer construction for high-power medium-voltage applications is described. The transformer includes a magnetic core, planar windings composed of one or more printed circuit boards (PCBs), an embedded cooling tube, and insulation layers that are substantially ceramic. Further embodiments include power transistors that switch current and control power flow through the device on each of the primary and secondary sides of the transformer to implement a compact power-transfer circuit. The transistors on each side of the transformer are attached to the same PCB and share the same insulation structures as at least one of the respective transformer windings. The embedded cooling tube could also be a heat pipe. Further

embodiments package the transistors as bare die in a transistor die assembly that reduces the thermal resistance between the bare die and the coolant.

[0009] In one example, a planar transformer includes a primary winding having one or more PCBs each including metallic traces. A secondary winding includes one or more PCBs having metallic traces. A ceramic electrical insulator electrically separates the primary and secondary windings. A magnetic core extends entirely around the windings and the electrical insulator.

[0010] In another example, a dual active bridge (DAB) module includes a planar transformer having a first winding and a second winding and a magnetic core. The first winding is formed as metallic traces on at least one first printed circuit board (PCB). The second winding is formed as metallic traces on at least one second PCB. The magnetic core being formed around at least a portion of the first and second PCBs. A first plurality of transistors is formed on at least one of the first PCBs to form a first stage. The first transistors are controlled to provide a first current through the first winding based on an input voltage to provide a second current in the second winding. A second plurality of transistors is formed on at least one of the second PCBs to form a second stage. The second transistors are controlled to provide an output voltage based on the second current.

[0011] In another example, a method for providing power to electric vehicles (EVs) at an EV charging station includes electrically coupling input stages of a respective plurality of dual active bridge (DAB) modules, each of the DAB modules being formed from a plurality of printed circuit boards (PCBs) and having a respective one of the input stages, a planar transformer comprising a primary winding and a secondary winding, and an output stage. The output stages of the respective DAB modules are electrically coupled, such that the DAB modules are arranged as a voltage converter circuit. An AC current is provided to the voltage converter circuit via the input stages of the DAB modules. Switches associated with the input stages of the respective DAB modules are controlled to provide a primary current through the primary winding of each of the planar transformers of each of the DAB modules to provide a secondary current in the second winding of each of the planar transformers of each of the DAB modules. Switches associated with the output stages of the respective DAB modules are controlled to provide an output voltage from each of the DAB modules based on the respective secondary current provided in the secondary winding of the planar transformer of each of the DAB modules. The output voltage is provided to each of a plurality of isolated EV chargers for the EV charging station.

[0012] In another example, a voltage converter circuit includes a plurality of dual active bridges (DABs). Each DAB has a first stage with a plurality of first switches, a second stage with a plurality of second switches, and a transformer having a first winding coupled to a least one of the first switches and a second winding coupled to at least one of the second switches. At least one of the first and second stages of a plurality of the DABs that is a proper subset of the DABs are arranged in parallel with respect to the first winding of the transformer in each of the respective plurality of the DABs, with the parallel first stages of the plurality of the DABs being arranged in series with respect to the first winding of the transformer of at least one other DAB of the DABs, or at least one of the first and second stages of a plurality of the DABs that is a proper subset of the DABs are arranged in series with respect to the second winding of the transformer in each of the respective plurality of the DABs, with the series second stages of the plurality of the DABs being arranged in parallel with respect to the second winding of the transformer of at least one other DAB of the DABs.

[0013] Other objects and advantages and a fuller understanding of the invention will be had from the following detailed description and the accompanying drawings.

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## Description

## BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. **1** is a schematic illustration of an example dual active bridge (DAB) module having a planar transformer in accordance with an aspect of the present invention.

[0015] FIG. **2** is a perspective view of a magnetic core of the transformer.

[0016] FIG. **3** is an exploded view of printed circuit boards (PCBs) for forming primary and secondary windings of the transformer.

[0017] FIG. **4**A is a perspective view of an insulating assembly for the transformer.

[0018] FIG. **4**B is an exploded view of the insulating assembly of FIG. **4**A.

[0019] FIG. **5** is a perspective view of a cooling structure for the transformer.

[0020] FIG. **6** is a section view taken along line **6-6** of FIG. **1**.

[0021] FIG. **7** is a top view of the DAB module with components removed.

[0022] FIG. **8** is a top view of the DAB module.

[0023] FIG. **9**A is a section view taken along line **9**A-**9**A of FIG. **8**.

[0024] FIG. **9**B is a section view taken along line **9**B-**9**B of FIG. **8**.

[0025] FIG. **10**A is a perspective view of a transistor assembly package of the DAB module.

[0026] FIG. **10**B is a top view of a portion of the DAB module with an array of transistors.

[0027] FIG. **10**C is a section view taken along line **10**C-**10**C of FIG. **10**B.

[0028] FIG. **11** is a schematic illustration of the DAB module enclosed in potting.

[0029] FIG. **12**A is a schematic illustration of a stand-alone planar transformer.

[0030] FIG. **12**B is a schematic illustration of the stand-alone transformer enclosed in potting and a case.

[0031] FIG. **13** is an example of a circuit diagram of a dual active bridge (DAB) circuit.

[0032] FIG. **14** is another example of a circuit diagram of a DAB circuit.

[0033] FIG. **15** is an example of a voltage converter circuit.

[0034] FIG. **16** is another example of an electric vehicle (EV) charging system.

## DETAILED DESCRIPTION

[0035] The present invention relates generally to transformers, and specifically to a planar, high power, medium voltage transformer and an associated DAB module, i.e., included in the same package assembly as a DAB module. FIGS. **1-10**C illustrate an example DAB module **501** having a planar transformer **10** and extending in a direction of length L in accordance with the present invention. It is appreciated that the planar transformer **10** can alternatively be provided in other, non-DAB modules or circuits or be stand-alone, as will be described.

[0036] As shown in FIG. **2**, the transformer **10** includes a core **20** formed from a pair of cooperating halves **22**, **24**. The halves **22**, **24** can be configured as E-cores or E-I-cores. Regardless, the core **20** is formed from a magnetic material, such as ferrite. In one example, the core **20** is a high frequency ferrite material, such as manganese zinc ferrite, nickel zinc ferrite or other ferrite suited for an operation frequency of about 20 kHz-200 kHz.

[0037] As shown, the halves **22**, **24** are E-cores that each includes a central support member **26**. The support members **26** from each half **22**, **24** engage one another and cooperate to define a pair of windows or passages **30**, **32** in the core **20**.

[0038] A pair of bobbins **36** encircle the respective support members **26** and are stacked atop one another so as to extend the entire collective height of the support members. Each bobbin **36** is generally tubular and includes a radially extending shelf or flange **38** that encircles the associated support member **26**. The bobbins **36** are made from an electrically insulating material, such as nylon, silicone, PPS, PPA, PBT or polyester. The bobbins **36** should also have a Comparative Tracking Index (CTI) of greater than 600V and high temperature capability.

[0039] The core **20** shown includes three associated pairs of halves **22**, **24** aligned along the length L. It will be appreciated, however, that more or fewer associated pairs of halves **22**, **24** can be provided. Regardless, a compression clip **28** is provided for each associated pair of halves **22**, **24** for biasing the halves towards one another. That said, three compression clips **28** are provided on the example core **20** shown. It will be appreciated that alternative/additional means could be used to secure the core **20**, e.g., adhesive, tape, other fasteners, etc.

[0040] Referring to FIG. **3**, the transformer **10** further includes a primary winding **40** and a secondary winding **42** formed from at least one secondary winding portion. In one example shown, a pair of secondary winding portions **42**A, **42**B is provided on opposite sides of the primary winding **40** and cooperate to define the secondary winding **42**. In any case, each winding **40**, **42** can be formed from one or more printed circuit boards (PCB) having either a first type **50** or a second type **70**. In the example shown, each winding **40**, **42** is formed from stacked PCBs **50**, **70**. It will be appreciated, however, that each PCB **50**, **70** could be formed as a single layer or multiple layers and that the PCBs forming each winding **40**, **42** could be single layers PCBs, multiple layer PCBs, combinations thereof, and/or combinations of PCBs having different number of layer(s) from one another within or between the windings **40**, **42**.

[0041] Regardless, the first PCB **50** includes a base **52** having an array of openings **54** extending through the thickness thereof. Tabs **55** extend outward from opposite sides of the base **52**. A projection **56** extends longitudinally away from the base **52**. An elongated opening **58** extends along the length of the projection **56** and passes entirely therethrough.

[0042] The second PCB **70** includes a base **72** and tabs **75** extending outward from opposite sides of the base. A projection **76** extends longitudinally away from the base **72**. An elongated opening **78** extends along the length of the projection **76** and passes entirely therethrough. Support members **80** extend outward from opposite sides of the tabs **75**.

[0043] Each PCB type **50**, **70** is formed from a dielectric material. Electrical conductors or traces (not shown) are etched into a side (the top side as shown) of each PCB **50**, **70** and generally encircle the respective opening **58**, **78**. The etched traces can extend one or more times around the respective openings **58**, **78**. The conductors can be made of copper or any other electrically conductive material and insulated with an epoxy-glass composite, such as FR-**4**. Fillers can be used in the insulation to enhance thermal conductivity.

[0044] As an example, the electrical conductors of at least one of the primary winding **40** and the secondary winding **42** can include multiple conductors arranged in parallel, such as to reduce eddy current losses. Additionally, for example, the conductor paths of the parallel conductors provided for the primary winding **40** and/or the secondary winding **42** can be transposed to link substantially equal magnetic flux of the planar transformer **10** to improve current sharing between the parallel conductors.

[0045] As shown, PCBs **50**, **70** are stacked on one another to form the primary winding **40**, with a pair of the PCBs **70** being sandwiched between pairs of the PCBs **50** in a six-layer construction. It will be appreciated that more or fewer of either PCBs **50**, **70** can be provided to form the primary winding **40**. The PCBs **50**, **70** are bonded to one another with solder or sintering to form the **42**A, **42**B primary winding **40**. Regardless, the openings **58**, **78** are aligned with one another and the bases **52**, **72** are aligned with one another. The PCBs **50**, **70** can be bonded to one another with solder or sintering to form the primary winding **40**.

[0046] Furthermore, as shown, PCBs **50**, **70** are stacked on one another to form each secondary winding portion **42**A, **42**B, with a pair of the PCBs **50** being positioned on one side of a PCB **70** in a three-layer construction. It will be appreciated that more or fewer of either PCB **50**, **70** can be provided to form the secondary winding portions **42**A, **42**B, but the secondary winding portions have the same configuration as one another. Regardless, the openings **58**, **78** are aligned with one another and the bases **52**, **72** are aligned with one another. The windings **40**, **42** are oriented such that the aligned bases **52**, **72** of the primary winding **40** is at an opposite end of the stack of PCBs than the aligned bases **52**, **72** on both secondary winding portions **42**A, **42**B.

[0047] Insulation formed as an electrical insulator or insulating assembly **100** is also provided (FIGS. **4**A-**4**B) for helping to electrically insulate the primary winding **40** from each secondary winding portion **42**A, **42**B. That said, components of the insulating assembly **100** are formed from an electrically insulating material, such as a ceramic. Example ceramic materials include, aluminum (III) oxide (Al.sub.2O.sub.3), aluminum nitride, silicon nitride, and boron nitride. As an example, the ceramic material(s) selected for the insulating assembly **100** can have dimensions that are suitable for providing a nominal operating potential between two of the electrical components, e.g., between the primary side and the secondary side of the transformer **10**, of at least 2 kV.

[0048] The insulating assembly **100** includes elongated first insulating sheets **102** each having an elongated opening **104** extending along the length thereof. Second insulating sheets **112** each have an elongated opening **114** extending along the length thereof. The second insulating sheets **112** are smaller in both length and width than the first insulating sheets **102**, i.e., the insulating sheets are different sizes. An edge insulator **120** has a generally U-shaped configuration and includes a base **122** and a pair of legs **124** extending therefrom parallel to one another. A pair of support rails **126**, **128** extends along the entire length of the edge insulator **120** and radially inward. The support rails **126**, **128** extend parallel to one another. An end cap insulator **140** has a generally box-shaped construction and includes a pair of C-shaped cover members **142**, **144** connected together by a common end wall **146**. A C-shaped projection **148** extends from the wall **146** into the interior between the cover members **142**, **144**.

[0049] When the insulating assembly **100** is assembled, two first insulating sheets **102** are positioned on opposite sides of the support rail **126** and abut both the support rail and the base **122**. A single second insulating sheet **112** is positioned between the first insulating sheets **102** and within the same plane as the support rail **126**. Similarly, two first insulating sheets **102** are positioned on opposite sides of the support rail **128** and abut both the support rail and the base **122**.

[0050] A single second insulating sheet **112** is positioned between the first insulating sheets **102** and within the same plane as the support rail **128**. This aligns all the openings **104**, **114** of the sheets **102**, **112** with one another. Due to this configuration, a three-piece or three-layer insulating member is associated with each rail **126**, **128** and constructed from a pair of first insulating layers **102** on opposite sides of a single second insulating layer **112**. The end cap insulator **140** extends over the free ends of the legs **124**.

[0051] Turning to FIG. **5**, a tubular cooling structure **160** extends from a first end **162** to a second end **164**. Straight portions **166** extend parallel to one another and are connected at their terminal ends by curved portions **168**. In one example, the straight portions **166** have flattened oval or generally rectangular cross-sections so as to increase the width (w) and decrease in thickness (t) compared to a rounder cross-section. The thickness (t) is on the order of about 0.05-1.0 mm, in particular about 0.1-0.30 mm. As will be discussed, the cooling structure **160** is configured to receive a cooling medium.

[0052] In any case, the portions **166**, **168** are configured to define multiple out-and-back passes of the cooling structure **160**, such as the pair of passes **172***a*, **172***b* shown. The passes **172***a*, **172***b* are connected end-to-end by a connecting portion **174**. The number of passes **172** is even to position the ends **162**, **164** on the same side of the cooling structure **160**. The cooling structure **160** is made of metal, such as stainless steel including a **300** series or austenitic stainless steel.

[0053] It will be appreciated that the cooling structure **160** could instead be formed as a metallic, e.g., stainless steel, heat pipe. In this configuration, the heat pipe has capillary structure and a working fluid, e.g., methanol, therein that boils at approximately the working temperature of the windings **40**, **42**. The heat pipe can have a rounded profile or the flattened profile exhibited by the cooling structure **160** shown. The heat pipe can have the same contour/shape as the cooling structure **160** shown, i.e., multiple passes **172***a*, **172***b*, or be configured with more or fewer passes. In one example, the heat pipe includes a single straight section **166**, and one or more heat pipes can extend parallel to one another through the respective passages **30**, **32** in order to span a desired percentage of the width of each passage.

[0054] Turning to FIGS. **6-7**, when the transformer **10** is assembled, the primary winding **40**, secondary winding portions **42**A, **42**B, insulating assembly **100**, and cooling structure **160** all extend through the passages **30**, **32** of the core **20** in a precisely tailored manner. To this end, the legs **124** of the edge insulator **120** extend through opposite lateral sides of the passages **30**, **32** and abut the interior of both halves **22**, **24** of the core **20**. In this manner, the support rails **126**, **128** are aligned with and extend parallel to the respective shelves **38** on the bobbins **36**. The insulating sheets **102**, **112** extend through the core **20** with the openings **104** receiving the bobbins **36** and the openings **114** receiving the shelf **38** on each bobbin **36**.

[0055] The primary winding **40** is positioned between both the shelves **38** on the bobbins **36** but also between the support rails **126**, **128** on both legs **124** due to the openings **58**, **78** in the PCBs **50**, **70**. This positions the primary winding **40** between a pair of the first insulating sheets **102**. A pair of second insulating sheets **112** is provided on each side of that stack, and a second pair of first insulating sheets **102** is then provided on each side of that stack. The insulating assembly **100**—in particular the sheets **102**, **112**—extends beyond the entire combined length of the respective secondary winding portions **42**A, **42**B and, thus the entire interface between the primary windings **40** and the secondary winding portions is electrically insulated.

[0056] The bases **52**, **72** of the primary winding **40** extend between the cover members **142**, **144** and within the end cap insulator **140**. The tabs **55**, **75** of the primary winding **40** extend laterally through the space between the cover members **142**, **144** to positions outside the end cap insulator **140**.

[0057] The secondary winding portions **42**A, **42**B, on the other hand, are provided outside the insulating assembly **100**. One secondary winding portion **42**A abuts the exterior of the outermost first insulating sheet **102** abutting the shelf **38** and the top (as shown) of the support rail **126**. The other secondary winding portion **42**B abuts the exterior of the outermost first insulating sheet **102** abutting the shelf **38** and the bottom (as shown) of the support rail **128**. The openings **58**, **78** allow the secondary winding portions **42**A, **42**B to exhibit this configuration. That said, the insulating assembly **100** and bobbins **36** cooperate to electrically insulate the primary and secondary windings **40**, **42** from one another and electrically insulate the windings from the core **20**. Furthermore, the insulating assembly **100** helps to increase the creepage distance between the primary winding **40** and each respective secondary winding portions **42**A, **42**B.

[0058] The cooling structures **160** are provided within the passages **30**, **32** and associated with each secondary winding portion **42**A, **42**B. In particular, one cooling structure **160** has one pass **172***a* extending through the passage **30** while the other pass **172***b* extends through the other passage **32**, with both passes being substantially in the same plane and atop the same, top (as shown) secondary winding portion **42**A. Similarly, the other cooling structure **160** has one pass **172***a* extending through the passage **30** while the other pass **172***b* extends through the other passage **32**, with both passes being substantially in the same plane and atop the same, bottom (as shown) secondary winding portion **42**B.

[0059] Due to this construction, a single cooling structure **160** provides a back-and-forth fluid path through one passage **30** and then another back-and-forth fluid path through the other passage **32**. It will be appreciated that the cooling structure **160** can be configured such that multiple, consecutive passes **172** extend through the same passage **30** and/or the passage **32**. Regardless, the ends **142**, **144** of the cooling structures **160** are positioned at the same end of the transformer **10**.

[0060] With that in mind, the straight portions **166** of each cooling structure **160** are aligned with and extend over the array of openings **54** in the secondary winding portions **42**A, **42**B. As shown, straight portions **166** are in close proximity to the openings **54** in the secondary winding portions **42**A, **42**B with no structure therebetween. The straight portions **166** of each cooling structure **160** are also aligned with the openings **54** in the primary winding **40**. The secondary winding portions **42**A, **42**B and insulating assembly **100**—specifically the second insulating sheets **112**—extend directly between the straight portions **166** and the openings **54** in the primary winding **40**.

[0061] The curved portions **168** extend into the end cap insulator **140** (FIGS. **8-9**B). In particular, the curved portions **168** of one cooling structure **160** are positioned on one side of the projection **148** and the curved portions **168** of the other cooling structure are positioned on the other side of the projection. In other words, the cooling structures **160** extend into the respective cover members **142**, **144**.

[0062] Each cooling structure **160**—specifically the straight portions **166**—can be electrically insulated from the associated secondary winding portions **42**A, **42**B with a thermal enhancement material (see **161** in FIG. **12**A). The thermal enhancement material **161** can be formed from, for example, thermally conductive silicone compounds or phase change materials. It will be appreciated that in lieu of [or in addition to] the thermal enhancement material **161** the PCBs **50** adjacent the cooling structure **160** can be configured such that the copper traces thereon are provided on the side of the PCB facing away from the cooling structure. In this sense, the PCB **50** itself [or rather its thickness] acts as an insulating layer between the cooling structure **160** and secondary winding portions **42**A, **42**B. In any case, the clips **28** help to load the cooling structure **160** against both insulating assembly **100** as well as the core **20**.

[0063] The DAB module **501** includes a transistor **250** provided in each of the aligned openings **54** within the primary winding **40** and each secondary winding portion **42**A, **42**B. As an example, the transistors **250** can each be provided as a metal-oxide semiconductor field effect transistor (MOSFET) having a transistor die assembly that includes one or more transistor devices. The transistors **250** can be fabricated as SiC MOSFETs, but could instead be any of a variety of different types of transistor devices, such as gallium nitride (GaN) devices, other silicon transistor devices, or silicon insulated-gate bipolar transistors (IGBTs). As an example, the transistors **250** can be either bare die components or can have a minimal surrounding package to allow a low inductance and a high thermal conductance connection to the insulating assembly **100**. Background information related to an example of the transistor **250** can be found in U.S. Pat. No. 7,119,447, the entirety of which is incorporated by reference.

[0064] Turning to FIGS. **10**A-**10**C, the transistor **250** shown includes an interposer **254** and a bare die **256** stacked atop one another. The bare die **256** can be formed from silicon carbide. The interposer **254** is positioned between the bare die **252** and a mounting surface, e.g., on one of the PCBs **50** and/or PCBs **70** for helping to transition the thermal coefficient of expansion and make mounting transistor **250** mounting more compatible with PCB manufacturing.

[0065] In the example of FIG. **10**A, the interposer **254** includes one or more electrical vias that extend through the interposer **254** to couple a gate terminal of the bare die **252** to a gate pad **255** and to couple a source terminal of the bare die **252** to a source pad **257** for providing electrical connectivity to the associated electrical traces of the mounting surface. The interposer **254** can be made of a material having a coefficient of expansion within about 5 ppm/K of the coefficient of expansion of the transistor **250**. Example materials for the interposer **254** include aluminum (III) oxide (Al.sub.2O.sub.3), aluminum nitride, and silicon nitride, and silicon dioxide.

[0066] The transistors **250** are incorporated into the primary and secondary windings **40**, **42** on the same PCBs **50**, **70** and using the single insulating assembly **100**. The transistors **250** associated with the primary side of the DAB module **501** are located next to the terminals of the primary winding **40**. The transistors **250** can have the same ground wall insulator to isolate the transistors **250** and connect to the transformer **10** on the same PCB **50**, **70** as at least one of the primary windings **40** on the transformer **10**.

[0067] A heat spreader **260** includes a base **262** and a pair of legs **264** that cooperate to define a recess **266** for receiving the interposer **254** and bare die **256**. In addition, the heat spreader **260** can correspond to a drain connection for the transistor **250**, such that the heat spreader is electrically connected to the drain terminal of the bare die **256**, e.g., on a surface of the bare die **256** opposite the interposer **254**. The heat spreader **260** is also electrically connected to the mounting surface via respective drain pads **265** arranged on the respective legs **264** and extending parallel to one another. That said, the drain pads **265** can be co-planar with one another. The heat spreader **260** can be made of a material having a coefficient of expansion within about **5** ppm/K of the coefficient of expansion of the transistor **250**. Example materials for the heat spreader **260** include molybdenum, tungsten, copper-tungsten, copper-molybdenum, and aluminum-graphite. That said, the components **252**, **254**, **260** can be connected to one another and to the PCB **50**, **70** with a sintered material made from silver or a silver alloy.

[0068] During operation, a cooling medium is supplied to the first ends **162** of the cooling structures **160**. The cooling medium can be, for example, water, a water mixture, glycol, a glycol mixture or combinations thereof. An ethylene glycol or propylene glycol additive can be included to help prevent freezing and fouling. In any case, the cooling medium flows through each pass **172***a*, **172***b* in succession, passing back-and-forth through each passage **30**, **32** in the core **22**.

[0069] As noted, the flattened portions **166** are aligned with and in close proximity to the transistors **250** in the secondary winding portions **42**A, **42**B. On the other hand, heat from the primary winding **40** passes through one or more the layers of its PCBs **50**, **70** and all the PCBs **50**, **70** in the secondary winding portions **42**A, **42**B, plus the insulating assembly **100**, before reaching the coolant in the cooling structures **160**. Consequently, a material with a thermal conductivity greater than the thermal conductivity (˜0.25 W/m-K) for flame retardant, woven glass-reinforced epoxy resin (FR4) is selected for the PCBs **50**, **70**.

[0070] The cooling structures shown and described herein provide a greater thermal conduction pathway connection to the windings and core due to their flattened profile. This profile also advantageously helps to reduce the space taken up by the cooling structures in the core passages while increasing the coolant Reynolds number, thereby increasing the fluid convection coefficient. Additionally, forming the cooling structure to pass back-and-forth through the same passage allows for both the tube and the coolant therein to be electrically conductive, which reduces construction costs and helps to improve operational reliability.

[0071] A series of spacers **180**, **181**, **182** are provided in the DAB module **501** for helping to protect the components therein and provide a more structurally sound and compact assembly. With this in mind, a pair of spacers **180** is provided for each secondary winding portion **42**A, **42**B on opposite sides of the core **20**, i.e., four total spacers **180** are provided. Each spacer **180** extends from the core **20** to the end of the straight portions **166** of the cooling structure **160** of that particular one of the secondary winding portions **42**A, **42**B. That said, the straight portions **166** are sandwiched between [but still spaced from] the outermost PCB **50** of that particular one of the secondary winding portions **42**A, **42**B and the spacer **180** associated therewith.

[0072] A spacer **181** (FIGS. **7** and **9**B) is provided between the curved portions **168** of each secondary winding portion **42**A, **42**B and the end wall **146** of the end cap insulator **140**. The spacers **181** extend to opposite sides of the projection **148** on the end wall **146**. A spacer **182** (FIG. **10**B) is also provided in the plane of the primary winding **40** and on the opposite side of the edge insulator **120** from the primary winding.

[0073] PCBs **190** are provided on both sides of the core **20** and extend along the length L of the DAB module **501** parallel to one another. The PCBs **190** receive the support members **80** extending from the tabs **75** on the primary winding **40** to secure the two together and laterally space the PCBs **190** from the secondary winding portions **42**A, **42**B. That said, capacitors **192** are connected to each PCB **190** on opposite lateral sides of the secondary winding portions **42**A, **42**B and opposite longitudinal sides of the core **20**.

[0074] The PCBs **190** each include openings for receiving a second core **200** that acts as a transformer of an isolated DCDC converter to provide control and gate-drive power from the secondary side to the primary side (or vice versa). The second cores **200** are made of a ferritic material and extends to opposite sides of the PCB **190**. Additional capacitors **192** are provided in the lateral space between the spacer **182** and each PCB **190**. Additional capacitors **194** can be provided on the tabs **55** of the primary winding **40** (FIG. **9**).

[0075] The transistors **250** associated with the primary side of the DAB module **501** are located next to the terminals of the primary winding **40**. The DAB module **501** can include operating power electronics, such as including gate drive power supplies, gate driver integrated circuits (ICs), voltage and current sense circuits, and switch control signals from a controller. Because there are active components on each side of the transformer isolation barrier formed by the transformer **10**, the DAB module **501** can include features for transferring power and logic signals across the isolation barrier. FIG. **8** shows one possible configuration that includes a long planar transformer core **20** to transfer power, and fiber optic links **210** between logic circuits to transfer logic signals across the isolation barrier.

[0076] Once the transformer **10** is assembled, the entire assembly is potted in a thermally conductive, electrically insulating potting compound **300** (FIG. **11**) that leaves the ends **162**, **164** of the cooling structures **160** exposed/accessible. Example materials for the potting compound **300** include nylon, silicone, PPS, PPA, PBT, epoxy, or polyester. In any case, the potting compound **300** should have a low durometer (<60 Shore 00) and a low un-cured viscosity (<5000 cPs), inserted under a vacuum.

[0077] Referring to FIG. **12**A, it will be appreciated that the transformer **10** can be modified to be a stand-alone assembly, i.e., one not implemented into the DAB module **501**. In such a configuration, the PCBs **50**, **70** are modified to omit the openings **54** for the transistors **250**. Additional DAB module **501** components, such as the PCBs **190**, capacitors **192**, core **200**, and fiber optic links **210** are also omitted. That said, the stand-alone transformer **10** can also be potted as shown in FIG. **12**B.

[0078] To this end, a case **310** cooperates with the potting **300** to increase the durability of the transformer **10**. The case **310** should be made of plastic, have a high temperature rating, and a high CTI greater than about 600V. The case **310** includes one or more compression clips **312** positioned along the length L of the potting **300** for providing a sustained, compressive force on the potting while also helping to mitigate any partial discharge in the potting that may occur during the life of the transformer **10**. The use of silicone for the potting **300**, in combination with the clips **312**, can help provide a self-healing function for the transformer **10**.

[0079] It will be appreciated that the case **310** can include one or more electronic couplings or connections **313** that enable the transfer of power from/between exterior of the potting **300** to the interior thereof. The connections **313** can be provided on both ends of the transformer **10**/DAB module **501**. That said, the connections **313** can also be provided on both ends of the case **310** shown in FIG. **11**.

[0080] The example of FIG. **13** illustrates an example of a circuit diagram of a DAB circuit **400** associated with the transformer **10** and DAB module **501** described herein. That said, the DAB circuit **400** is demonstrated as a high-frequency DC-to-DC converter that includes a transformer **402** with a series inductor, a first stage **404** that includes a full active bridge, and a second stage **406** that likewise includes a full active bridge. In the example of FIG. **13**, the transformer **402** is demonstrated as including an inductor in series with the primary winding. The inductor can be a separate inductor, or can correspond to a leakage inductance of the transformer **402** if the leakage inductance is sufficient to not require a separate inductor.

[0081] The first stage **404** is demonstrated as an input stage in which the switches of the full active bridge are switched to provide a current through the primary winding of the transformer **402** from an input voltage V.sub.IN across an input capacitor C.sub.1. Similarly, second stage **406** is demonstrated as an output stage in which the switches of the full active bridge are switched to provide an induced current through the secondary winding of the transformer **402** to provide an output voltage V.sub.OUT across an output capacitor C.sub.2. The DAB circuit **400** can be provided in a high-power voltage converter to provide for bidirectional power flow, low modulation complexity, case of resonant conversion, e.g., because it can be operated at fixed **50**% duty cycle, and soft-switching, and high conversion efficiency.

[0082] The DAB circuit **400** is demonstrated in the example of FIG. **13** and described above as unidirectional. However, it is to be noted that the description of the DAB circuit **400** as converting a DC voltage at the input stage via the primary winding to provide a DC voltage at the output stage via a secondary winding is one example implementation regarding the orientation of the DAB circuit **400**. As another example, the DAB circuit **400** can operate bidirectionally, such that the input stage/primary and output stage/secondary could be reversed. The potential bidirectional operation of the DAB circuit **400** can likewise be applicable to the further examples described below.

[0083] As an example, the transformer **10** can be implemented as the transformer **402** in the DAB circuit **400** to achieve high-power voltage conversion in a compact form-factor, such as the DAB module **501**, as described herein. The example of FIG. **14** illustrates another example of a circuit diagram of a DAB circuit **500**. The DAB circuit **500** is demonstrated similar to the DAB circuit **400** in the example of FIG. **13**, and thus includes a transformer **502**, a first stage **504** that includes a full active bridge, and a second stage **506** that likewise includes a full active bridge. However, the first stage **504** also includes an additional full active bridge, such that the DAB circuit **500** can provide AC-DC conversion. Similar to as described above, the transformer **10** can be implemented as the transformer **502** in the DAB circuit **500** to achieve high-power voltage conversion in a compact form-factor. Furthermore, the DAB circuit **500** can correspond to the DAB module **501** described above in the example of FIG. **1**.

[0084] For example, the switches of the first and second stages **504** and **506** can correspond to the transistors **250** described above, and the capacitors C.sub.1 and C.sub.2 can correspond to the capacitors **190** and **192** described above. In addition, the DAB circuit **500** includes a set of controls for operating the DAB circuit **500**. The controls include a master controller **508**, a set of isolated communication logic **510**, low-side logic **512**, and high-side logic **514**. As an example, the master controller **508** and/or the communication logic **510** can be external to the DAB circuit **500**, and thus external to the DAB module **501**, e.g., to control all DAB modules **501** in a given set of DAB modules, as described in greater detail below. The isolated communication logic **510**, secondary-side logic **512**, and primary-side logic **514** can be included in the DAB module **501**. For example, the fiber-optic link **210** of the DAB module **501** can be implemented to provide logic signals between the primary-side logic **514** and the secondary-side logic **512**.

[0085] To achieve conversion of the medium amplitude AC voltage to the high amplitude DC voltage, a group of DAB circuits **500** can be electrically combined to form a voltage converter circuit. The example of FIG. **15** demonstrates a voltage converter circuit **700** that includes a plurality of DAB circuits **702**, demonstrated as twelve in the example of FIG. **15**, that provide voltage conversion for an EV charging station **704**. As an example, each of the DAB circuits **702** can correspond to the DAB circuit **500** described above to collectively convert a medium amplitude AC voltage, such as at least 5 kVAC (e.g., 7.5 kVAC), provided from an AC input **706** to a high amplitude DC voltage, such as at least 750 VDC (e.g., 1000 VDC), provided to the EV charging station **704**. Each of the DAB circuits **702** includes an input (primary) stage **708** and an output (secondary) stage **710** that are interconnected by a high-frequency transformer **712**, e.g., the planar transformer **10** described above.

[0086] In the example of FIG. **15**, the primary windings of the transformers **712** of each of the DAB circuits **702** are connected in series via the input stage **708**, and the secondary windings of the transformers **712** of each of the DAB circuits **702** are connected in parallel via the output stage **710**. As a result, the voltage converter circuit **700** exhibits a series-in parallel-out architecture. By placing the primary windings of the transformer **712** in series and the secondary windings of the transformer **712** in parallel, a large voltage step-down can be achieved using approximately 1:1 turns ratio of the transformers **712**. It is noted that turns ratios other than 1:1 can instead be implemented, e.g., between approximately 0.5:1 and 2:1. This general topology allows for isolated voltage conversion without the use of large, e.g., 50 or 60 Hz, transformers. A higher transformer frequency allows for the use of smaller transformers, such as the transformer **10**. Similar to as described above, while the term "primary" refers to the AC voltage side and the term "secondary" refers to the DC side, the configuration of the voltage converter circuit **700** can operate bidirectionally, such that the primary and secondary could be reversed in practice.

[0087] In the example of FIG. **15**, the voltage converter circuit **700** also includes an inductor L.sub.IN in series with the AC input **706**, thereby rendering the front end of the voltage converter circuit **700** to act as an active boost rectifier providing the DC voltage on the capacitor C.sub.1 in the input stage **708** of each of the DAB circuits **702**. The combination of the AC-DC DAB circuits **702** having the inputs connected through the primary winding of each of the transformers **712** in series provides for a multilevel converter input stage which allows multiple lower voltage transistors to be used as the switches of the input stages **708** to accommodate a high voltage input from the AC input source **706**.

[0088] For example, as described above, the transistors of the input and/or output stages **708** and **712** can be arranged as the transistors **250** of the DAB module **501**. The series inductor L.sub.IN for the series connected modules can make the input stages **708** of the voltage converter circuit **700** operate as a cascaded active boost rectifier. While the single inductor L.sub.IN is demonstrated in the example of FIG. **15** in series with the series-connected input stages **708**, an individual inductor can instead be included in the input stage **708** of each of the DAB circuits **702**, and thus included in the DAB module **501**, such as to provide for a more modular construction of the voltage converter circuit **700**.

[0089] The voltage converter circuit **700** can each be configured in a variety of different ways to provide AC-DC conversion, DC-DC conversion, DC-AC conversion, or AC-AC conversion. For example, an additional full active bridge can be provided to the output of each of the DAB circuits **700** to provide an AC output voltage, e.g., 600 VAC, collectively. As another example, the secondary windings of the transformers **712** in the output stages **710** of the DAB circuits **700** may be coupled together in different ways in a given one of the voltage converter circuits **802**. For example, for an even number of DAB modules **700**, pairs of the output stages **710** of respective DAB circuits **702** can be arranged in series instead of parallel to provide twice the output voltage, e.g., 2 kV, at half the output current than the fully parallel connection shown in the example of FIG. **15**.

[0090] Similarly, pairs of the input stages **708** of respective DAB circuits **702** can be arranged in parallel instead of series to provide twice the input current over a uniform AC voltage for each pair than the fully series connection shown in the example of FIG. **15**. The voltage converter circuit **700** can thus be configured with any variation of AC or DC input or output, and any combination of series and parallel couplings of the input stages **708** and the output stages **710** to affect the amplitude of the output voltage V.sub.OUT.

[0091] The example of FIG. **16** demonstrates another example of an EV charging system **800**. The EV charging system **800** includes three separate phases of AC power, Phases A through C, that are provided to three respective voltage converter circuits **802**. Each of the voltage converter circuits **802** can correspond to the voltage converter circuit **700**, and can thus each include a plurality of DAB circuits **500**, e.g., that each correspond to the DAB module **501**. Accordingly, each of the voltage converter circuits **802** can implement the same DAB circuits **500** for each of the voltage converter circuits **802**, thereby providing for a simple modular design of each of the voltage converter circuits **802**.

[0092] In the example of FIG. **16**, the three separate phases of AC power are provided from AC power sources **804** to provide **13** kVAC three-phase power to the three voltage converter circuits **802**, such that each of the voltage converter circuits **802** can generate 1000 VDC. As an example, the 13 kVAC three-phase power can be connected in a wye configuration to provide the 7.5 kVAC to each of the voltage converter circuits **802**, thus reducing the number of DAB circuits **500** in each of the respective voltage converter circuits **802**. Alternatively, the 13 kVAC three-phase power can be connected in a delta configuration to provide the 13 kVAC to each of the voltage converter

circuits **802**.
[0093] The EV charging system **800** includes a master controller **806** that can draw sinusoidal current in phase with the source phase voltages to achieve near-unity power factor, i.e., power correction (PFC). As an example, the master controller **806** can regulate the output voltage on the capacitors C.sub.2 of each of the DAB circuits **500**, while simultaneously maintaining an approximately equal voltage across all the capacitors Ci in each of the DAB circuits **500**. Therefore, the master controller **806** can provide active voltage balancing in providing the output voltage to each of a plurality of isolated EV chargers for the EV charging system **800**.
[0094] For example, the master controller **806** can control the switches of the first and second stages **504** and **506** between the capacitors C.sub.1 and C.sub.2 of each of the DAB circuits **500** to perform the sinusoidal current control in conjunction with regulating the voltage on the respective capacitor C.sub.2. In the example of FIG. **16**, the three phases of the AC power **804** are balanced, and all of the capacitors C.sub.2 of each of the DAB circuits **500** are connected in parallel across the three respective power converter circuits **802**. Therefore, the three balanced phases of the AC power **804** can result in minimization of the total amount of capacitance of the capacitors C.sub.2. The cascaded AC/DC front end stages of each of the DAB circuits **500** in each of the voltage converter circuits **802** act to maintain approximately balanced and equal voltages on the capacitors C.sub.1 by directing the input AC current in or out of each of the capacitors C.sub.1 as needed for proper regulation.
[0095] What have been described above are examples of the present invention. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the present invention, but one of ordinary skill in the art will recognize that many further combinations and permutations of the present invention are possible. Accordingly, the present invention is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims.

## Claims

**1**. A protective enclosure for electrical components, comprising: potting material encasing the electrical components; a case covering the potting material; and at least one elastic component extending over the case for applying a compressive load to the potting material.
**2**. The enclosure recited in claim 1, wherein the potting material has a durometer less than about 60 Shore 00.
**3**. The enclosure recited in claim 1, wherein the potting material comprises silicone.
**4**. The enclosure recited in claim 1, wherein the potting material is a thermally conductive, electrically insulating material.
**5**. The enclosure recited in claim 1, wherein the case comprises a plastic having a comparative tracking index of greater than about 600V.
**6**. The enclosure recited in claim 1, wherein the electrical components include a planar transformer.
**7**. The enclosure recited in claim 1, wherein a nominal operating potential between two of the electrical components is greater than 2 kV.
**8**. The enclosure recited in claim 1, wherein the at least one elastic component comprises multiple compression clips arranged along the length of the case.
**9**. The enclosure recited in claim 8, wherein the potting material and the compression clips cooperate to provide self-healing.
**10**. The enclosure recited in claim 1, wherein the case completely covers the potting material.
**11**. The enclosure recited in claim 10, wherein the case includes electrical connections for enabling the transfer of power between the interior and exterior of the potting material.
**12**. A protective enclosure for a planar transformer having cooling structure, comprising: potting material encasing the planar transformer; a plastic case covering the potting material; and

compression clips extending over the case and arranged along the length thereof for applying a compressive load to the potting material.

**13**. The enclosure recited in claim 12, wherein the cooling structure extends through the case.

**14**. The enclosure recited in claim 12, wherein the potting material has a durometer less than about 60 Shore 00.

**15**. The enclosure recited in claim 12, wherein the potting material comprises silicone.

**16**. The enclosure recited in claim 12, wherein the case completely covers the potting material.

**17**. The enclosure recited in claim 16, wherein the case includes electrical connections for enabling the transfer of power between the interior and exterior of the potting material.